

## Research Paper

## Social media big data reveals how mobility reshapes human environmental exposure inequality



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## HIGHLIGHTS

- We compared residence-based and mobility-based measures of green exposure across 317 Chinese cities.
- Residence-based measures significantly overestimate green exposure and its inequality.
- When using mobility-based measures, the disparity between large and small cities decreases.
- Differences in green exposure between high and low-income groups narrowed.
- This is the first large-scale study incorporating human mobility data in green exposure assessment.

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## ABSTRACT

For a long time, environmental exposure is considered closely related to people's socioeconomic status (SES). Assessing this hypothesis seemed straightforward, as we could calculate the accessibility of green spaces around the residences of different socioeconomic groups, naturally concluding that high-SES residents enjoy higher green exposure. However, this assessment method relies on static residence-based evaluations without considering people's daily mobility. In this study, we utilized over 100 million real activity locations and 1 million street view images to measure the mobility-based green exposure of approximately 20,000 users. We found that compared to the mobility-based green exposure, traditional residence-based assessments overestimate green exposure by 11.26% and green exposure inequality by as much as 54.44% (with the Gini coefficient decreasing from 0.5163 to 0.3343). The primary driver of changes in green exposure inequality is the reduction of within-group disparities. Further, after considering users' mobility behavior, although high-income groups still experience higher levels and equality of green exposure compared to low-income groups, the relative gap in green exposure narrowed to 4.27%. This change is even more pronounced between cities, with the relative difference in green exposure between small and large cities shrinking from 32.70% to 16.83%. This study marks the first large-scale application of human mobility in environmental exposure research, significantly advancing our understanding of environmental exposure and inequality. The findings challenge long-standing conclusions in green exposure inequality studies, demonstrating that urban green exposure equity is higher than traditionally perceived. These results are also likely to be applicable to research on other similar environmental exposure issues.

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## 1. Introduction

Green exposure refers to the degree to which people interact with or are exposed to natural environments and green spaces. Substantial evidence indicates that green exposure in urban areas is closely linked to human health (Barboza et al., 2021). As cities serve as the primary venues of human activity, ensuring fair access to green spaces for people of different ages, races, ethnicities, and socioeconomic statuses has become a key challenge in the urbanization process. This goal aligns with several United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-being), SDG 10 (Reduced Inequalities), and SDG 11 (Sustainable Cities and Communities). Evidence suggests that the degree of green exposure inequality in cities of developing countries is twice that of developed countries (Chen et al., 2022). This significant gap highlights the urgent need for both accurate assessment of disparities and targeted interventions to promote more equitable green exposure. These patterns underscore the importance of ensuring inclusive, equitable, and sustainable green exposure for all social groups in order to mitigate broader social inequalities.

Our understanding of human green exposure levels and their inequality has improved in recent years, yet important methodological limitations remain. The mainstream assessment method represents green exposure by calculating the green space coverage around people's homes. Substantial evidence has demonstrated significant disparities in environmental exposure across different population groups, potentially leading to varying levels of health risks (Jbaily et al., 2022; Zhang & Kwan, 2025). However, accurately measuring an individual's environmental exposure remains a critical challenge. People spend their time not only at home but also at various locations they travel to or visit routinely, such as shopping malls, workplaces, and recreational sites. If human mobility is not taken into account when considering green exposure, it may lead to overestimation or underestimation of people's actual green exposure (Kwan, 2012; Perchoux et al., 2013). It means that ignoring people's daily mobility and exposure to out-of-home and travel contexts may lead to biased or even erroneous conclusions (Chaix et al., 2017; Kwan, 2018a). But evaluating this hypothesis is challenging because traditional methods of obtaining data on individual mobility require many participants and dedicated portable devices, resulting in a lack of large-scale and individual-level evidence.

Additionally, given the limitations of the residence-based approach, it is crucial to reassess green exposure inequalities. It is also important to examine whether the patterns observed from residence-based measures still hold once human mobility is taken into account. Evidence from the United States and Europe indicates that there are significant disparities in green exposure among people of different socioeconomic statuses (SES) and ethnicities (Wüstemann et al., 2017; Spotswood et al., 2021). In particular, ethnic minorities and groups with low SES often face lower green exposure (Jbaily et al., 2022; Kim et al., 2023; Testi et al., 2024). However, most evidence is mostly based on residence-based exposure assessments. These measures mainly reveal that these groups have poorer green space provision around their residences. There is little evidence to show that there are also significant inequalities in mobility-based exposures among different groups.

Given the importance of considering human mobility in environmental exposure assessment, some scholars have attempted to apply user-based behavioral experiments and spatiotemporal big data approaches to evaluate environmental exposure under population mobility (Zhang & Kwan, 2025). The first method mainly uses GPS tracking or small-scale travel surveys to examine mobility-based environmental exposure. But it is typically limited to small samples. And it is difficult to provide systematic evidence across a large number of cities and a large number of individuals. The latter approach also has several limitations. For example, approaches based on GPS or phone signaling data are often vulnerable to the modifiable areal unit problem (MAUP) when locations are aggregated into arbitrary spatial units.

In comparison, social media check-in data provide an alternative

representation of human behavior that is relatively easy to access and contains less sensitive personal information. Moreover, check-in locations are precisely geocoded at the point level, which helps mitigate MAUP-related issues. Since becoming available, such data have been widely used in various domains of urban science. These characteristics make social media check-in data particularly suitable for constructing large-scale, individual-level, mobility-based exposure measures across many cities. Existing residence-based methods also suffer from spatial misrepresentation bias, as they implicitly assume that environmental exposure occurs exclusively within residential neighborhoods while ignoring the complex spatiotemporal patterns of human activity spaces. The neglect of human activity spaces violates the basic principles of time geography and environmental exposure assessment, where an individual's cumulative exposure represents the integrated effect of all environments encountered throughout their daily activity patterns (Kwan, 2012, 2018; Cai & Kwan, 2024).

To address these challenges, this study proposes a large-scale mobility-based green exposure calculation method using social media users' activities. The key motivation is straightforward: people do not experience environmental conditions only at their homes, but also at multiple locations they visit in their daily lives. Cumulative green exposure thus arises from all places individuals visit—such as home, workplaces, shopping areas, and recreational sites—rather than from residential neighborhoods alone. By integrating street view, social media, and matched real estate price data (for assessing the SES level) from the same time period, we can more accurately assess individual green exposure in urban areas. The Neighborhood Effect Averaging Problem (NEAP) provides the theoretical mechanism explaining why mobility-based measurements may show different inequality patterns compared to residence-based assessments. When people move around, their exposures tend to average out across diverse urban environments rather than being confined to potentially segregated residential neighborhoods. This framework enables us to systematically compare residence-based and mobility-based green exposure across a large number of cities and social groups, and to quantify how much human mobility changes both the overall level of exposure and the degree of inequality.

We selected more than 300 cities in China to evaluate how human mobility impacts green exposure levels and inequalities compared to traditional residence-based assessments, at a scale that has rarely been examined in previous mobility-based exposure studies. Specifically, we seek to answer the following three questions:

RQ1: How does incorporating human mobility change the measured levels of individual green exposure?

RQ2: Does considering mobility alleviate or exacerbate green exposure inequality compared to static residence-based assessments?

RQ3: Are the mobility-induced changes in green exposure and its inequality significantly different among various cities and SES groups?

## 2. Methods

### 2.1. Experimental design

We conducted a national-scale analysis utilizing 110 million social media records across China to uncover the mobility-induced disparities in green exposure inequality. We constructed the activity locations of 15 million users based on their social media check-in locations. Subsequently, we determined the type of each check-in location, extracting real estate type as the user's residence. We then identified users' SES based on property data by dividing housing prices in each city according to percentiles, thereby classifying users into high, medium, and low SES groups (Nilforoshan et al., 2023). We applied very stringent screening criteria, including that the user had published more than 5 blog posts and had publicly shared activities at residential-type locations. The final sample included 21,640 users, of which 20,905 had gender attributes, from over 300 cities. Subsequently, we used street view images to

calculate users' green exposure levels. Two green exposure measures were calculated for each user: the first, called residence-based green exposure, represents the green exposure levels around the user's residence; the second, called mobility-based green exposure, represents the average green exposure at the user's activity locations, with the latter considering more comprehensive. The calculated method was shown in the Fig. 1.

## 2.2. Construction of mobility data and estimation of users' socioeconomic status

We constructed user mobility activity locations using public check-in data from Sina Weibo, which is China's largest social media platform. The dataset has over 110 million check-in records of 15 million users throughout 2023, including anonymized user IDs, timestamps, user-shared tweet texts, and geographic coordinates. Additionally, it contains publicly available user attributes such as age, gender, and education level. To ensure data quality and meaningful trajectory reconstruction, we processed these data through several steps. We built a semantic classification model to categorize visited locations into over ten types, including residential, commercial, industrial, and transportation. Our method is based on all locations visited by users, with weights determined by the frequency of user posts. For example, if a user's visit records are A, A, B, C, D, then the user's green exposure would be the sum of (Exposure-A + Exposure-A + Exposure-B + Exposure-C + Exposure-D)/5 as the representation of mobility-based exposure.

We retained only users who had visited residential locations during nighttime, marking these locations as their residences, and subsequently used internet property data to label users' socioeconomic attributes. To ensure data reliability, we filtered the dataset to retain approximately 20,000 users who had continuous check-in activities and had publicly shared their residential locations. Furthermore, estimating users' SES is crucial. We collected average housing price data (yuan/square meter) for the communities where users' residences were located, sourced from Lianjia, China's largest online real estate platform. We used a quantile

method to label users' SES (Guzman et al., 2026). For example, if user A's community housing price is in the top 25% of their city's average housing prices, we labeled them as high SES. If user B's residential housing price is in the bottom 25% of their city, we labeled them as low SES, while users in the middle 50% were labeled as middle SES. This method has been widely used in previous studies and is considered a feasible approach for large-scale approximation of user attributes (Nilforoshan et al., 2023; Guzman et al., 2026).

## 2.3. Assessment of inequality

To assess inequality in residence and mobility-based environmental exposure among different groups, we employed the Gini coefficient, a widely used distribution statistic. The Gini coefficient ranges from 0 to 1, where 0 represents perfect equality and 1 represents maximum inequality. The calculation of the Gini coefficient is straightforward. Assuming we calculate the environmental exposure values for 100 individuals in group A, we sort the exposure values from low to high and calculate the cumulative proportion of the population and the cumulative proportion of total exposure. We then plot these proportions as a Lorenz curve and calculate the area between this curve and the line with a slope of 1. The Gini coefficient is twice this area. By comparing Gini coefficients under different scenarios, we can quantify the impact of mobility on environmental exposure for different groups. The formula for the Gini coefficient (G) is:

$$G = \frac{1}{2n^2\mu} \sum_i \sum_j |x_i - x_j|$$

where  $n$  is the number of individuals,  $\mu$  is the average exposure, and  $x_i$  and  $x_j$  are the exposure values for individuals  $i$  and  $j$ , respectively. We calculated Gini coefficients for both residence and mobility-based exposure for each group. This allowed us to compare inequality levels not only between socioeconomic groups but also between different exposure scenarios.



Fig. 1. Schematic diagram of mobility-based exposure and residence-based exposure calculation. (Note: 1d is . adapted from Kim and Kwan, 2021b)

#### 2.4. Verification of the Neighborhood Effect Averaging Problem (NEAP)

The Neighborhood Effect Averaging Problem (NEAP) is a central concept in our research, explaining the differences between individual-level mobility-based and residence-based environmental exposures in a study area (Huang and Kwan, 2022; Kim and Kwan, 2021a; Kim and Kwan, 2021b; Kwan, 2018b). The NEAP has been supported by the results of over a dozen studies. It posits that when people's daily mobility is taken into account, their actual exposure to environmental factors tends to converge towards the average level within the same city. Individuals residing in high-exposure areas may actually experience lower exposure (downward averaging), while those living in low-exposure areas may experience higher exposure (upward averaging). To examine the NEAP, we conducted the following statistical analyses. First, we calculated the mean mobility-based green exposure (EM) and mean residence-based green exposure (ER) for all study groups. We defined the NEAP effect as:

$$\text{NEAP effect} = (\text{EM} - \text{ER}) / \text{ER}$$

This formula quantifies the relative difference between EM as a percentage of the ER. And then we calculated the variance for both EM ( $\text{Var}_m$ ) and ER ( $\text{Var}_r$ ):

$$\text{Var}_m = \sigma_m^2$$

$$\text{Var}_r = \sigma_r^2$$

where  $\sigma$  represents the standard deviation. We also conducted paired sample t-tests to compare the differences between EM and ER:

$$t = (\text{EM} - \text{ER}) / (S_d / \sqrt{n})$$

Where  $S_d$  is the standard deviation of the differences, and  $n$  is the sample size. We considered NEAP to exist if the following conditions were met:

NEAP effect  $\neq 0$ , and if NEAP effect  $> 0$  means upward averaging, NEAP effect  $< 0$  means downward averaging.  $\text{Var}_m < \text{Var}_r$ , suggesting more uniform exposure levels across different locations for individuals when mobility is considered.  $p < 0.05$  in the  $t$ -test, indicating a statistically significant difference between mean mobility-based and residence-based exposures.

This methodological approach allows us to examine whether individual exposure levels tend to converge towards the mean of the study area when accounting for daily mobility. It provides a framework to assess how mobility affects environmental exposure patterns, potentially revealing situations where individuals with initially low exposure levels may experience increased actual exposure, while those with initially high exposure levels may see decreased actual exposure. Based on these statistical tests, we can explain the causes of changes in green exposure inequality, thereby further revealing the limitations of residence-based environmental exposure.

#### 2.5. Calculation of green exposure using Baidu street view imagery (Baidu SVI)

The most common method for objectively assessing green exposure is based on remote sensing data, such as the Normalized Difference Vegetation Index (NDVI) and land cover geographic datasets. Consistent with findings from Zhang et al. (2021), the NDVI and green view index (GVI) demonstrate a significant positive correlation. However, studies have found discrepancies between objectively measured green exposures based on satellite remote sensing images and people's perceived green exposures (Wang et al. 2019). Remote sensing images can only provide an overhead view, making it difficult to capture information at eye level and on the vertical attributes of green spaces (such as wall vegetation and tree height). To overcome this limitation, we use street view images to characterize people's visual green exposure levels (Middel et al., 2019; Zhang et al., 2024). Compared to satellite assessments, street view images are closer to the actual visual experience at the street level and are more relevant to human health (Helbich, 2019; Lu et al., 2018).

To be specific, we utilized Baidu Street View (BSV) imagery, which

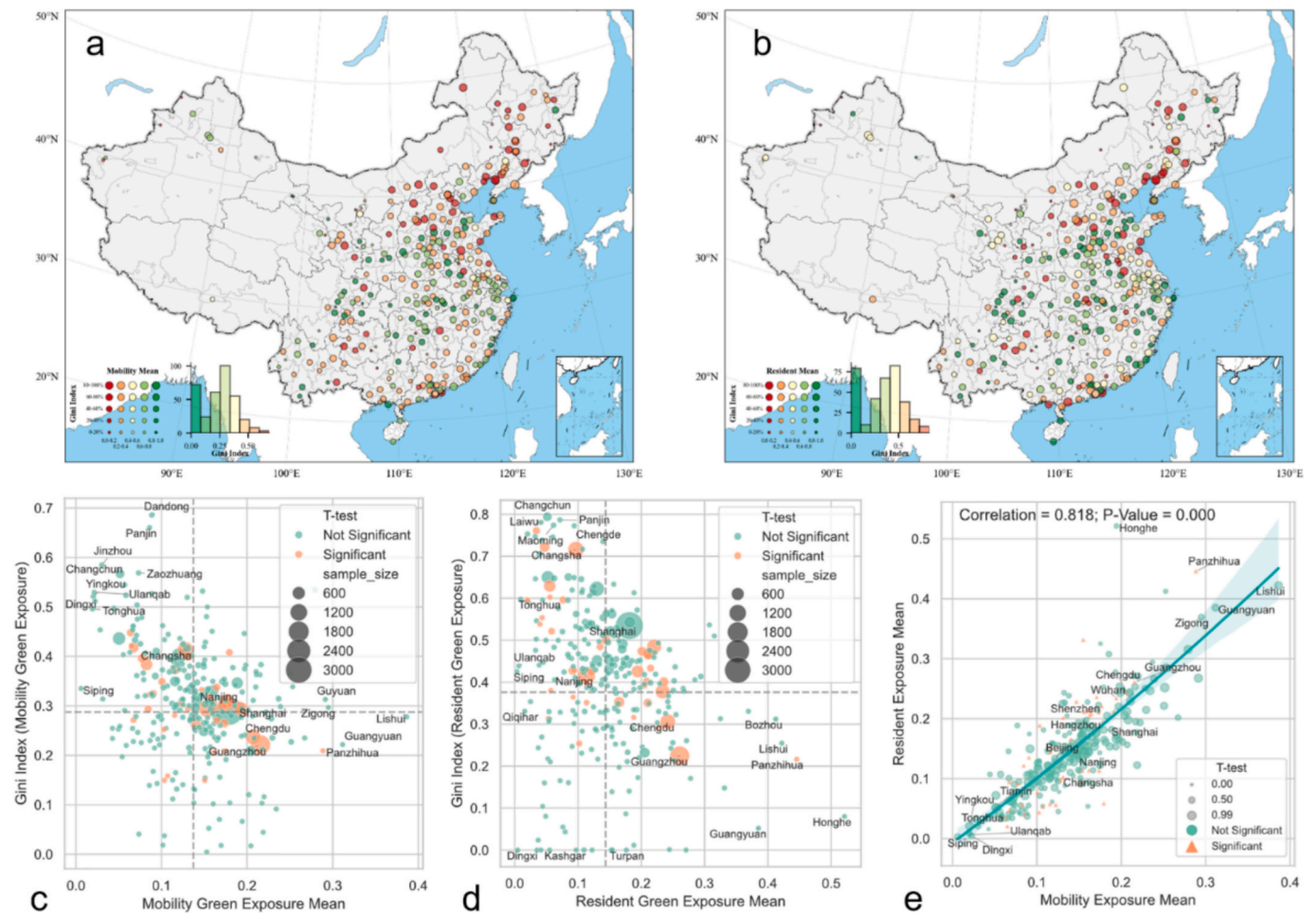
provides panoramic views of urban environments. We collected about 1 million BSV images for each user's residential location and activity points, capturing the street-level visual environment. The image collection process involved querying the Baidu Maps API with the geographic coordinates of each location, retrieving four images per location oriented at  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$ , and  $270^\circ$  to ensure comprehensive coverage. To extract green information from these images, we employed a deep learning-based semantic segmentation model. Specifically, we used a pre-trained DeepLabv3+ model with a ResNet101 backbone, fine-tuned on the cityscape dataset. This model categorizes each pixel in the image into one of several classes, including vegetation, buildings, sky, and other urban elements (Fig. S6). We focused on the vegetation class to calculate the GVI, defined as the ratio of green pixels (vegetation element) to the total number of pixels in the image. For each location, we computed the average GVI across the four directional images to obtain a single representative value. This method allows for an objective and scalable assessment of street-level greenery, capturing the green exposure that people experience in their daily environments. By applying this approach to both residential and activity locations, we are able to compare static (residential) and dynamic (mobility-based) green exposures, providing insights into how human mobility influences individuals' green exposure equality.

### 3. Results

Fig. 2 reveals the differences in green exposure and inequality (Gini coefficient, see Methods, The Assessment of Inequality) between users' residence-based and mobility-based measurements at the city level. Fig. 2a uses bubbles to represent the green exposure and inequality levels of cities without considering human mobility. Each bubble represents a city, with larger bubble sizes indicating higher levels of inequality. Bubbles that are more red indicate lower levels of green exposure, while those that are more green indicate higher levels of green exposure. Fig. 2b uses the same visualization method to show the green exposure and inequality levels of cities based on mobility-based exposures. The subplots in Fig. 2a and 2b display the distribution of Gini indices across different cities. It can be observed that the inequality levels in cities are noticeably lower. Fig. 2c and 2d show the exposure values on the horizontal axis, with the mobility-based and residence-based exposure values represented respectively, while the vertical axis shows the degree of inequality in the two types of exposure. The scatter plot for the mobility-based exposure is more concentrated. Fig. 2e shows the regression fit results for those two green exposures, with a correlation coefficient of 0.818.

Overall, users' mobility-based green exposure is on average about 11.26% lower than their residence-based green exposure (RQ1, indicating systematic overestimation by traditional methods). Although we can observe a fairly strong positive correlation between mobility-based exposure and residence-based exposure of users in different cities (Fig. S1). Further, residence-based exposure overestimates green exposure inequality by approximately 54.44% (average green exposure decreased from 0.1688 to 0.1498, and the average degree of inequality decreased from 0.5163 to 0.3343) (RQ2, demonstrating that traditional assessments significantly overestimate environmental inequality). Changes in Table 1 exhibit systematic variation, with larger cities showing greater exposure reduction (13.38% vs. 1.63% in small cities) and high-SES groups experiencing proportionally larger decreases (12.97% vs. 11.30% for low-SES). (RQ3, indicating differential mobility benefits across socioeconomic strata).

Specifically, in Wuhan, the average mobility-based green exposure is about 0.1946, while the average residence-based green exposure value is about 0.2340, with the Gini coefficient (degree of inequality) decreasing from 0.3777 to 0.2890. In Beijing, these differences are even more pronounced, with the average mobility-based green exposure at about 0.1210, while the average residence-based green exposure is about 0.1318, and the Gini coefficient decreases from 0.6205 to 0.3909. In



**Fig. 2.** Comparison of Changes in Residence-based Green Exposure and Mobility-based Green Exposure in 317 Chinese Cities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

**Table 1**  
Differences in residence-based and mobility-based green exposures and inequality levels for different individual comparison groups.

| Group       | Mobility Mean ± Std | Residence Mean ± Std | Sample Size | Exposure Difference | Mobility Gini ± Std | Residence Gini ± Std |
|-------------|---------------------|----------------------|-------------|---------------------|---------------------|----------------------|
| Overall     | 0.1498 ± 0.0945     | 0.1688 ± 0.1634      | 20,905      | −11.26%             | 0.3343 ± 0.0018     | 0.5163 ± 0.0020      |
| medium SES  | 0.1466 ± 0.0935     | 0.1636 ± 0.1584      | 10,559      | −10.39%             | 0.3367 ± 0.0025     | 0.5184 ± 0.0027      |
| Low SES     | 0.1499 ± 0.0979     | 0.1690 ± 0.1651      | 5298        | −11.30%             | 0.3451 ± 0.0036     | 0.5201 ± 0.0040      |
| high SES    | 0.1563 ± 0.0925     | 0.1796 ± 0.1713      | 5048        | −12.97%             | 0.3170 ± 0.0034     | 0.5063 ± 0.0038      |
| Small City  | 0.1390 ± 0.0990     | 0.1413 ± 0.1465      | 2733        | −1.63%              | 0.3742 ± 0.0053     | 0.5377 ± 0.0055      |
| Big City    | 0.1624 ± 0.0976     | 0.1875 ± 0.1739      | 10,676      | −13.38%             | 0.3182 ± 0.0024     | 0.4994 ± 0.0026      |
| Medium City | 0.1334 ± 0.0899     | 0.1485 ± 0.1443      | 5526        | −10.17%             | 0.3591 ± 0.0035     | 0.5228 ± 0.0039      |
| Gender: m   | 0.1475 ± 0.0882     | 0.1694 ± 0.1634      | 7554        | −12.93%             | 0.3172 ± 0.0030     | 0.5134 ± 0.0032      |
| Gender: f   | 0.1512 ± 0.0979     | 0.1686 ± 0.1634      | 13,292      | −10.32%             | 0.3435 ± 0.0023     | 0.5177 ± 0.0025      |

Note: The classification of small and large cities is based on the definition of the government, which based on the combination of city population and GDP. The exposure and Gini values in the table are followed by ± 1 standard deviation. The exposure variance is the square of the Std. Users who did not share their gender were not included in the gender comparison group.

contrast, for Lanzhou, residence-based assessments underestimate green exposure, with mobility-based measures showing higher average exposure (from 0.1235 to 0.1310) and lower inequality (Gini coefficient from 0.3592 to 0.6136). However, regardless of the city, the degree of green exposure inequality is lower when using mobility-based measures. This finding highlights the limitations of traditional residence-based assessments that do not take into account of people's daily mobility. And it underscores the importance of incorporating human movement patterns when evaluating environmental exposure.

To further explain the reasons for the overestimation of inequality and elucidate the extent to which it occurs, we constructed different

control groups based on population characteristics and conducted analyses at the individual, location and city levels. We established eight individual comparison groups based on users' gender, SES, and city scale. Subsequently, we calculated the discrepancies between these two measures of green exposure for each group. The results are presented in Table 1 and Fig. S3.

For the high SES group, the average green exposure decreased from 0.1796 to 0.1563, with inequality levels dropping from 0.5063 to 0.3170 after mobility-based measures were used. The low SES group saw their average green exposure decrease from 0.1688 to 0.1498, with inequality levels falling from 0.5201 to 0.3451 after mobility-based

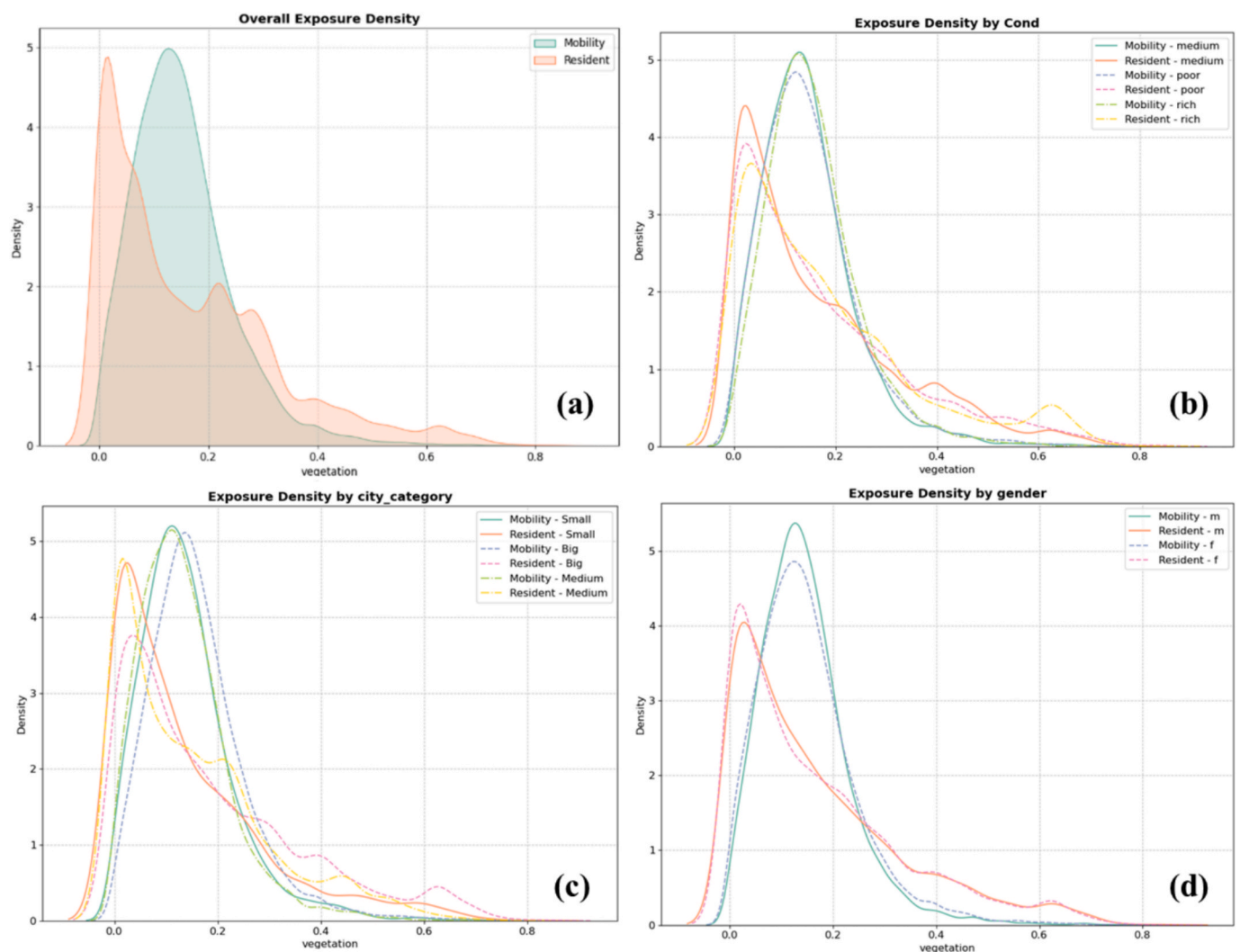
measures were used. The relative difference in green exposure between the two groups (high SES and low SES) decreased from 6.27% to 4.27%, while the absolute difference decreased from 1.05% to 0.64%. These changes are even more pronounced across different cities. Residents of large cities experienced a reduction in green exposure from 0.1875 to 0.1624, with inequality levels declining from 0.4994 to 0.3182 after mobility-based measures were used. The small city group saw their green exposure decrease from 0.1413 to 0.1390, with inequality levels dropping from 0.5377 to 0.3742. The relative difference in green exposure between the two groups (large cities and small cities) decreased from 32.70% to 16.83%, while the absolute difference decreased from 4.62% to 2.34%. For the gender groups, there are no significant differences between males and females, but the exposure inequality for both groups is lower after mobility-based measures were used.

From these comparison groups, we can observe that residents of large cities and those with high SES enjoy more abundant and equitable green exposure. Large cities have more population, providing governments with more economic resources and tax revenues, which often lead to better municipal greening. Similar significant differences have been observed in related studies (Nesbitt et al., 2019). In addition, an intriguing pattern emerges in our analysis. Initially, large cities and high SES groups show slightly more equitable green exposure distribution than other groups. However, when accounting for human mobility,

these inequality gaps widen significantly. The Gini coefficient difference between large and small cities increases from 7.12% to 17.60%, while the gap between high and low SES groups expands from 2.65% to 8.86%. This suggests that while mobility reduces overall green exposure inequality within cities, it paradoxically amplifies disparities between different cities and socioeconomic groups.

Additionally, we use the NEAP effect (see Methods for the NEAP Verification) to do an exposure difference test (All cities' test results can be found in the supplementary Table S2). The overall exposure difference (Table 1) is -11.26%, indicating green exposure in residential areas is generally higher than in other daily activity locations. The differences are most pronounced in the large city group, reaching -13.38%, with the standard deviation of mobility-based exposure (0.0976) significantly lower than that of residence-based exposure (0.1739). This decreased standard deviation (variance) implies a potential mitigation of environmental exposure inequalities across urban spaces due to daily human mobility. The difference is least evident in the small city group (-1.63%).

Fig. 3 presents the frequency plots of mobility-based and residence-based green exposure for different comparison groups. Fig. 3a represents the overall green exposure; Fig. 3b divides the population into high, middle, and low SES groups; Fig. 3c divides the population into residents of large, medium-sized, and small cities; Fig. 3d divides the population into male and female groups; The shapes of the curves are more



**Fig. 3.** The frequency plots of individual mobility-based and residence-based green exposure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

compressed and concentrated, moving inward at both the high and low ends.

The distribution curves of mobility-based exposure (Fig. 3a) exhibited a narrower range and a higher central peak. This concentration is further evidenced by lower variance (standard deviation decreases from 0.1634 to 0.0945), suggesting that human mobility tends to cause green exposure to converge towards the mean value. This convergence indirectly leads to lower levels of green exposure inequality. Fig. 3b–d illustrate the changes in green exposure frequency curves for other groups. Notably, a consistent trend emerges across all comparison groups: the mobility-based green exposure distributions are more uniform and concentrated compared to their residence-based methods, with exposure levels in both the high and low ends moving inward towards the average. This pattern holds true regardless of the group under consideration in Fig. 3.

Although we can see the changes in green exposure after considering mobility from Fig. 3, it remains unclear which types of non-residential locations have positive or negative effects. To explain the reasons for the changes in green exposure at the location level, we plotted the changes in green exposure when residents visit non-residential types of locations. In Fig. 4, the green points indicate locations where the levels of greenery are higher than the average levels of greenery in residential areas, while the red points indicate the opposite. For example, Shanghai's average residence-based green exposure is 0.2366, while the average mobility-based green exposure is 0.1907. When residents visit the green points, they experience an upward averaging effect, as these locations have higher levels of greenery than their residential areas. Conversely, when they visit the red points, they undergo a downward averaging effect, as these areas have lower levels of greenery than their residential surroundings. As a result of these downward and upward averaging trends, individual exposures tend to converge towards the mean of the study area. We plotted the differences between the residence-based and mobility-based green exposures of all activity locations (Fig. 4b). It can be seen that in the city center with high-density buildings, many locations have lower green exposure than the average residence-based exposure. Frequent human activities in this location lead to a green exposure reduction and downward averaging. This phenomenon can also be observed in multiple cities such as Chengdu, Beijing, Guangzhou, and others (Fig. 4a, 4b, 4d, and 4f). Although much of the evidence above points towards mobility reducing green exposure, this effect is not invariable. Fig. 4c and 4e show the opposite directions,

residents can access better green exposures outside the residence. The final mobility-based green exposure reaches an average exposure level. The supplementary Fig. S4 illustrates the disparities in green exposure changes across various location types.

Finally, in Fig. 5, we plotted the temporal differences in green exposures. It shows the 24-hour variations in green exposure for different SES groups in representative cities. Red lines represent high SES groups, while blue lines represent low SES groups. The green histogram indicates the degree of green exposure at different times. Overall, the exposure disparities between cities are substantially larger than those between different SES groups within cities. The daytime green exposure is generally higher than nighttime exposure (2–6 AM), which closely correlates with human activity patterns. Furthermore, during morning and evening rush hours, the differences in green exposure between different groups are minimal, reflecting the commuting patterns during these periods. These findings further support our previous observations: while human mobility generally reduces green exposure inequality within cities, the differences in urban development levels remain the primary factor driving green exposure disparities.

#### 4. Discussion and conclusion

Green exposure in urban areas is closely linked to human health, and the pursuit of equal access to greenery is a common goal in fostering public well-being. Equitable and adequate urban green exposure is closely related to multiple United Nations Sustainable Development Goals (SDGs), particularly SDG 3: Good Health and Well-being, SDG 11: Sustainable Cities and Communities, and SDG 10: Reduced Inequalities. In this study, we uncovered a research gap in traditional residence-based green exposure assessments, which could significantly affect the reliability of environmental exposure inequality studies. Considering that people's green exposure not only occurs near their residential locations but also around their travel routes and places they routinely visit (Kwan, 2013), we discovered that traditional residence-based methods significantly overestimate or underestimate residents' actual green exposure, resulting in considerable overestimation of green exposure. And we discovered that traditional methods overestimate green exposure inequality by approximately 54.44%, a bias stemming from fundamental differences between residential location sampling and comprehensive activity space assessment. Because individuals access diverse environmental conditions through daily mobility patterns, which moderate

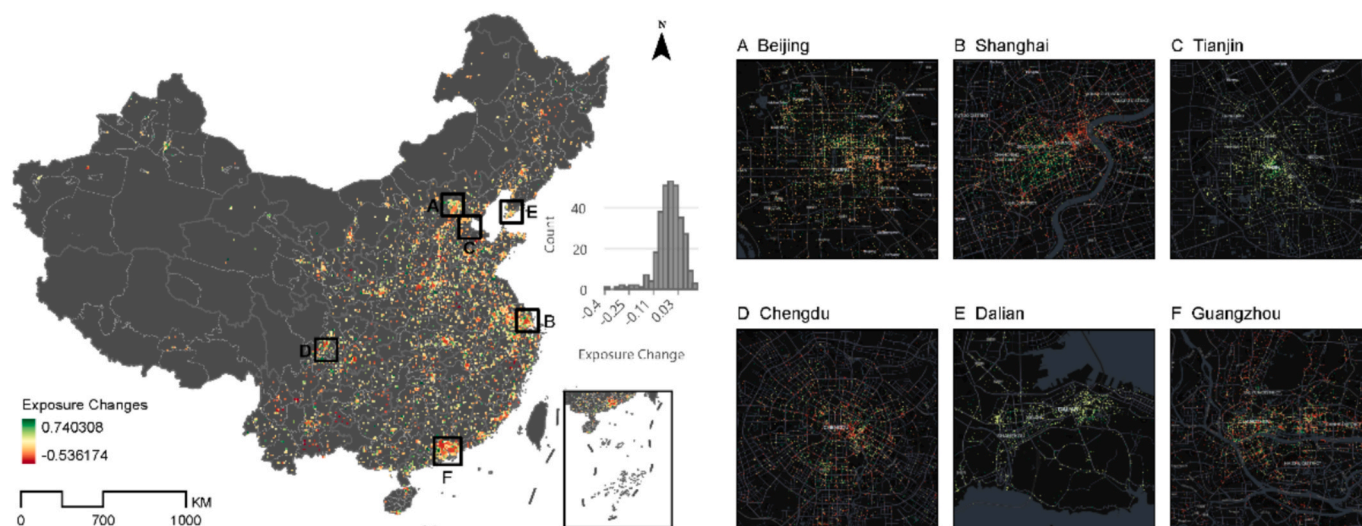


Fig. 4. Green exposure changes in non-residential locations (Visualization of greenery levels at activity locations implemented using [kepler.gl](https://kepler.gl). A detailed visualization dashboard of green exposure differences for 317 cities, totaling 206,694 activity locations, is shared on figshare. The green points indicate non-residential locations where greenery levels are higher than the average level of residential greenery, while the red points indicate the opposite.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 5.** Temporal patterns of green exposure across socioeconomic groups in different cities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

residential environmental disadvantages, residence-based methods actually capture spatial segregation patterns rather than actual exposure experiences. More importantly, while traditional methods overestimate overall inequality within cities, they simultaneously underestimate actual disparities between different cities and socioeconomic groups, revealing a dual bias that has profound implications for environmental justice.

This work makes three significant contributions. First, comprehensive individual-level spatiotemporal granularity (**response to RQ1**): This extensive study covers over 20,000 users across mainland China, analyzing 206,518 activity locations. It significantly advances environmental exposure research by overcoming traditional limitations in data scale, spatiotemporal resolution, and individual characteristics captured. Second, we systematically assess the changes in inequality (**response to RQ2**): We reveal that traditional residence-based methods overestimate green exposure inequality by approximately 54.44%. This overestimation is primarily attributed to the smaller variance of mobility-based green exposures compared to residence-based exposures. Third, challenge conventional perspectives (**response to RQ3**): our research findings contrast with numerous studies suggesting that urban expansion exacerbates inequality (Wu et al., 2024; Gao et al., 2024; Nilforoshan et al., 2023; Li et al., 2024). We found that inequality levels in large cities are not as high as previously observed (Wang et al., 2018). Due to the concentration of resources and population, the relatively uniform distribution of urban infrastructure actually leads to more equitable green exposure among residents in large cities. This may explain the recent intense urbanization waves in developing countries like China, with large populations migrating from small to large cities (Zhang et al., 2022). The result of residents “voting with their feet” demonstrates the enormous attraction of cities’ superior conditions, higher efficiency, and fairer social environments. This finding has important implications for understanding urbanization patterns and challenges the prevailing narrative that urban growth necessarily leads

to increased environmental inequality.

However, our study also has certain limitations. These include potential errors in inferring people’s SES, discontinuities in social media data, and biases arising from users selectively posting their activity locations. Due to the lack of large-scale official data, this paper’s measurement of SES is based on housing prices (yuan/m<sup>2</sup>) from Lianjia.com data for residential communities, which may also introduce bias. We also acknowledge the limitations inherent in social media mobility data, including demographic bias toward younger, educated populations and temporal sampling constraints. Nevertheless, compared to traditional mobile phone signaling and GPS data, social media data offers distinct advantages, including more comprehensive user information and higher accuracy of location (Pappalardo et al., 2023). In addition, we adopted a weighting scheme based on the posting frequency of social media users. Despite the inherent uncertain geographic context problem (Kwan, 2012), our analysis is the first to reveal the impact of human mobility on green exposure at a cross-city scale, providing more comprehensive evidence on green exposure inequality. Nevertheless, future research could integrate actual dwell time data from sources such as mobile phone signaling or GPS tracking, or develop hybrid weighting approaches that combine posting frequency with estimated dwell time to further enhance the accuracy of exposure assessment.

In addition, this study provides several practical implications for urban planning and green space management. Planning and management authorities should shift from focusing solely on “residence-based green provision” to “actual green exposure based on daily activity spaces,” incorporating commuting corridors, employment centers, and high-frequency activity destinations into green space planning and performance evaluation frameworks. More importantly, the “hidden” vulnerable groups whose residential neighborhoods appear to have sufficient greenery but whose daily activity spaces lack adequate green exposure should be identified and considered. Our research not only challenges traditional green exposure assessment methods and inherent

views on inequality levels, but the conclusions may also be applicable to other environmental exposures (air pollution, noise pollution). However, further research is still needed to provide evidence on this. Future research should extend this framework to other environmental exposures and diverse urban contexts to establish generalizability across different socio-spatial settings.

### CRediT authorship contribution statement

**Yan Zhang:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Mei-Po Kwan:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Haoran Ma:** Visualization. **Libo Fang:** Validation, Data curation. **Zeqiang Chen:** Software, Data curation. **Nengcheng Chen:** Formal analysis, Data curation.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.landurbplan.2026.105600>.

### Data availability

Data will be made available on request.

### References

- Barboza, E. P., Cirach, M., Khomenko, S., et al. (2021). Green space and mortality in European cities: A health impact assessment study. *The Lancet Planetary Health*, 5(10), e718–e730.
- Cai, J., & Kwan, M. P. (2024). The universal neighborhood effect averaging in mobility-dependent environmental exposures. *Environmental Science & Technology*, 58(45), 20030–20039.
- Chaix, B., Duncan, D., Vallée, J., Vernez-Moudon, A., Benmarhnia, T., & Kestens, Y. (2017). The “residential” effect fallacy in neighborhood and health studies: Formal definition, empirical identification, and correction. *Epidemiology*, 28(6), 789–797.
- Chen, B., Wu, S., Song, Y., et al. (2022). Contrasting inequality in human exposure to greenspace between cities of Global North and Global south. *Nature Communications*, 13(1), 4636.
- Gao, S., Chen, Y., Chen, D., et al. (2024). Urbanization-induced warming amplifies population exposure to compound heatwaves but narrows exposure inequality between global North and south cities. *npj Climate and Atmospheric Science*, 7(1), 154.
- Guzman, L. A., Cantillo-Garcia, V. A., Criado, A., Sarmiento, O. L., & Moncada, C. (2026). What street do you like? Exploring the subjective perception of public spaces in unequal and segregated cities. *Cities*, 168, Article 106460.
- Helbich, M. (2019). Using deep learning to examine street view green and blue spaces and their associations with geriatric depression in Beijing, China. *Environment International*, 126, 107–117.
- Huang, J., & Kwan, M.-P. (2022). Uncertainties in the assessment of COVID-19 risk: A study of people's exposure to high-risk environments using individual-level activity data. *Annals of the American Association of Geographers*, 112(4), 968–987.
- Kim, K., Joyce, B. T., Nannini, D. R., et al. (2023). Inequalities in urban greenness and epigenetic aging: Different associations by race and neighborhood socioeconomic status[J]. *Science Advances*, 9(26), Article ead8140.
- Jbaily, A., Zhou, X., Liu, J., et al. (2022). Air pollution exposure disparities across US population and income groups. *Nature*, 601(7892), 228–233.
- Kim, J., & Kwan, M.-P. (2021a). How neighborhood effect averaging may affect assessment of individual exposures to air pollution: A study of ozone exposures in Los Angeles. *Annals of the American Association of Geographers*, 111(1), 121–140.
- Kim, J., & Kwan, M.-P. (2021b). Assessment of sociodemographic disparities in environmental exposure might be erroneous due to neighborhood effect averaging: Implications for environmental inequality research. *Environmental Research*, 195, Article 110519.
- Kwan, M. P. (2012). The uncertain geographic context problem. *Annals of the Association of American Geographers*, 102(5), 958–968.
- Kwan, M. P. (2013). Beyond space (as we knew it): Toward temporally integrated geographies of segregation, health, and accessibility: Space–time integration in geography and GIScience. *Annals of the Association of American Geographers*, 103(5), 1078–1086.
- Kwan, M. P. (2018a). The limits of the neighborhood effect: Contextual uncertainties in geographic, environmental health, and social science research. *Annals of the American Association of Geographers*, 108(6), 1482–1490.
- Kwan, M. P. (2018b). The neighborhood effect averaging problem (NEAP): An elusive confounder of the neighborhood effect. *International Journal of Environmental Research and Public Health*, 15(9), 1841.
- Li, Y., Svenning, J. C., Zhou, W., et al. (2024). Green spaces provide substantial but unequal urban cooling globally. *Nature Communications*, 15(1), 7108.
- Lu, Y., Sarkar, C., & Xiao, Y. (2018). The effect of street-level greenery on walking behavior: Evidence from Hong Kong. *Social Science & Medicine*, 208, 41–49.
- Middel, A., et al. (2019). Urban form and composition of street canyons: A human-centric big data and deep learning approach. *Landscape and Urban Planning*, 183, 122–132.
- Nesbitt, L., Meitner, M. J., Girling, C., et al. (2019). Who has access to urban vegetation? a spatial analysis of distributional green equity in 10 US cities. *Landscape and Urban Planning*, 181, 51–79.
- Nilforoshan, H., Looi, W., Pierson, E., et al. (2023). Human mobility networks reveal increased segregation in large cities. *Nature*, 624(7992), 586–592.
- Pappalardo, L., Manley, E., Sekara, V., et al. (2023). Future directions in human mobility science. *Nature Computational Science*, 3(7), 588–600.
- Perchoux, C., Chaix, B., Cummins, S., & Kestens, Y. (2013). Conceptualization and measurement of environmental exposure in epidemiology: Accounting for activity space related to daily mobility. *Health & Place*, 21, 86–93.
- Spotswood, E. N., Benjamin, M., Stoneburner, L., et al. (2021). Nature inequity and higher COVID-19 case rates in less-green neighbourhoods in the United States. *Nature Sustainability*, 4(12), 1092–1098.
- Testi, I., Wang, A., Paul, S., et al. (2024). Big mobility data reveals hyperlocal air pollution exposure disparities in the Bronx. *New York. Nature Cities*, 1(8), 512–521.
- Wang, Q., Phillips, N. E., Small, M. L., et al. (2018). Urban mobility and neighborhood isolation in America's 50 largest cities. *Proceedings of the National Academy of Sciences*, 115(30), 7735–7740.
- Wang, R., Helbich, M., Yao, Y., et al. (2019). Urban greenery and mental wellbeing in adults: Cross-sectional mediation analyses on multiple pathways across different greenery measures. *Environmental research*, 176, Article 108535.
- Wu, S., Chen, B., An, J., et al. (2024). The interplay of cloud cover and 3D urban structures reduces human access to sunlight. *Nature Cities*, 1–9.
- Wüstemann, H., Kalisch, D., & Kolbe, J. (2017). Access to urban green space and environmental inequalities in Germany. *Landscape and Urban Planning*, 164, 124–131.
- Zhang, Y., Chen, N., Du, W., et al. (2021). Multi-source sensor based urban habitat and resident health sensing: A case study of Wuhan, China. *Building and Environment*, 198, Article 107883.
- Zhang, Y., Kwan, M. P., & Ma, H. (2024). Sensing noise exposure and its inequality based on noise complaint data through vision-language hybrid method. *Applied Geography*, 171, Article 103369.
- Zhang, Y., Zheng, X., Helbich, M., et al. (2022). City2vec: Urban knowledge discovery based on population mobile network. *Sustainable Cities and Society*, 85, Article 104000.
- Zhang, Y., & Kwan, M. P. (2025). Considering human mobility is essential for accurate environmental exposure assessment. *Science Bulletin*.